

MANUFACTURING METHODS AND MATERIAL EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device may be manufactured using techniques selected to achieve durability, comfort, sealing integrity, and scalable production while maintaining functional equivalence across implementations.

The insertable measurement section may be produced using overmolding processes in which compliant outer materials are molded around internal structural backbones and embedded sensing components to improve sealing, comfort, and sensor protection.

Backbone structures may include flexible polymers, reinforced elastomeric cores, segmented supports, or semi-rigid internal frames configured to maintain sensor alignment while permitting controlled flexibility during insertion and use.

Inflatable compliance regions may be manufactured using bonded elastomeric membranes, welded polymer films, molded silicone chambers, or layered composite structures designed to withstand repeated expansion cycles. Bonding approaches may include thermal welding, adhesive bonding, insert molding, or co-molding techniques.

Sensing components may be integrated using flexible printed circuit boards, rigid-flex assemblies, encapsulated sensor modules, or distributed microelectronic packages embedded within compliant materials. Encapsulation may include conformal coatings, potting compounds, or sealed protective layers to resist fluid exposure.

Distal sensing windows may utilize optically or acoustically transmissive materials configured to preserve measurement performance while protecting underlying sensors. Window structures may be flush-mounted, recessed, replaceable, or integrally molded into the distal tip.

The handle portion may be fabricated using injection molding, additive manufacturing, machining, or hybrid fabrication methods. Internal cavities may support batteries, pumps, processors, and communication modules while maintaining ergonomic handling characteristics.

Assembly processes may include modular subassemblies in which insertable sections, inflatable components, and electronic control modules are produced separately and combined into a unified device to improve manufacturability and serviceability.

Quality-control procedures may include automated testing of pressure integrity, calibration verification, inflation response behavior, communication performance, and electrical safety prior to distribution.

Accordingly, the disclosed manufacturing embodiments enable production ranging from simplified consumer configurations to advanced research or clinical-support instruments while maintaining consistent sensing performance, safety characteristics, and long-term durability.